Fluid Dynamics and Bernoulli's Principle

SYNOPSIS

Note 1 was fluid standing still. Everything there came from one idea — the fluid above you has weight. Now we set the fluid moving, and two new ideas take over, both of them conservation laws you already know, rewritten for something that flows.

Mass is conserved, so whatever goes into a pipe has to come out of it. Squeeze the pipe and the fluid has no choice but to speed up — that is the **continuity principle**.

Energy is conserved, so a fluid that speeds up has to pay for the extra kinetic energy out of something else. What it pays with is *pressure*. That is **Bernoulli's equation**, and it is the work–energy theorem with the terms divided through by volume.

Put the two together and you get the result that sounds wrong the first time you hear it: **where a fluid moves faster, its pressure is lower.** It is why a wing lifts, why a shower curtain gets sucked inward, and why the narrow part of a pipe is the part at risk of collapsing.

BEFORE YOU READ ON — PREDICT

A pipe carrying water narrows to half its width and then opens out again. In the narrow section, is the water pressure **higher** than in the wide section, **lower**, or the **same**? Commit to one in your guided notes, with a reason, before you read on.

PERFORMANCE EXPECTATIONS

State the continuity principle and use $A_1v_1 = A_2v_2$. Explain where Bernoulli's equation comes from, term by term, and say what each term is the energy of. Use it to compare two points on a streamline. Explain why faster flow means lower pressure, and apply that to a wing.

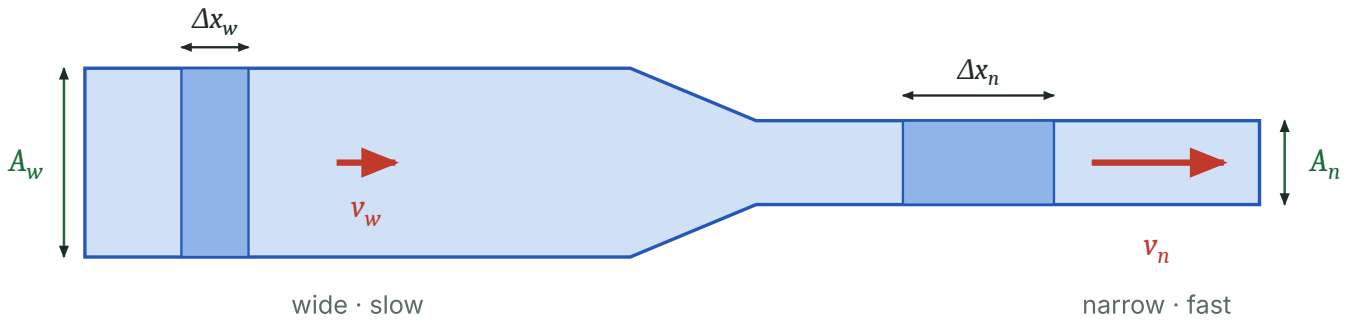
For PHY 151 — with calculus: Get Bernoulli's equation by integrating Euler's equation along a streamline, see exactly which assumptions the integral needs, and recover Torricelli's result for a draining tank as a special case.

WATCH · THE NOTE WORKED THROUGH

Recording to come. Print the guided notes (link at the bottom of this page), watch, and fill them in as you go.

Continuity — the fluid has to go somewhere

Water goes into one end of a pipe and comes out of the other. None of it is created on the way through and none of it disappears, so in any given second, **the amount that enters has to equal the amount that leaves**. That is mass conservation, and for a fluid of constant density it is volume conservation too. The product Av is the **volume flow rate**, Q, in m^3/s — litres per second by another name — and continuity says Q is the same everywhere along the pipe. Now watch what that forces on a pipe that changes width.



the two shaded slugs hold the same volume — and they take the same time to pass

Drawn as a duct of fixed depth, so the **height on the page is the area**. The wide section has **2.25×** the area of the narrow one, and every other ratio in the picture is the same 2.25: the narrow slug is 2.25× as long, and the narrow arrow is 2.25× as long. The two shaded slugs therefore have equal volume, which is the whole content of the continuity principle.

1. In a time Δt the slug in the wide section travels Δx_w and the slug in the narrow section travels Δx_n. (A) Write the statement that the two shaded volumes are equal. (B) Divide it by Δt and say what

each ratio has become. (C) Which is larger here, v_w or v_n, and why?

2. Water flows at 1.2 m/s through a pipe of cross-sectional area 0.040 m². (A) Find Q, in m³/s and in litres per second. (B) The pipe narrows to 0.015 m². What is the speed there?

3. You put your thumb over the end of a hose and the water reaches further. Of A, v and Q, say which changed, which did not, and why.

i) (A) $A_w \Delta x_w = A_n \Delta x_n$. (B) dividing by Δt gives $\boldsymbol{A_w v_w = A_n v_n}$, because $\Delta x/\Delta t$ *is* the speed. (C) $\boldsymbol{v_n}$

read Av = constant as a see-saw: $A_w > A_n$ forces $v_w < v_n$. Narrow means fast

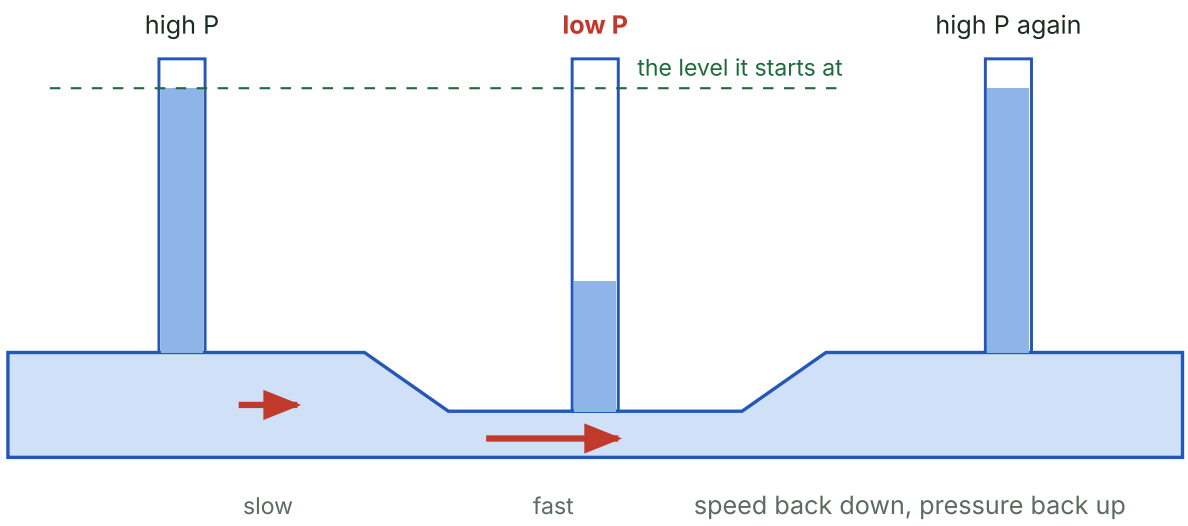
ii) (A) $Q = Av = (0.040)(1.2) =$ **0.048 m³/s = 48 L/s**. (B) $v = Q/A = 0.048/0.015 =$ **3.2 m/s** the area fell to 0.375 of its value and the speed rose by 1/0.375 = 2.67, which is the same number

iii) A fell, v rose, Q did not change — the tap sets Q, not your thumb. The water leaves faster, so it travels further before gravity brings it down

Fast flow is low-pressure flow

Here is where intuition goes wrong. Squeezing a pipe *sounds* like it should raise the pressure inside the squeeze — you are cramming the same water through a smaller hole. The opposite happens. The narrow section is the **low**-pressure section.

The reason is energy. The fluid arrives at the throat moving slowly and leaves it moving fast, so its kinetic energy went up. Nothing pushed it but the fluid behind, so that energy had to come from somewhere, and the only account left to draw on is pressure. The fluid spends pressure to buy speed.



Standpipes read the pressure in the duct below them: the taller the column, the higher the pressure. The column over the **throat** is the short one. Note also that the pressure **recovers** on the far side — the fluid gets its pressure back when it slows down again, which is what tells you nothing was lost, only traded.

4. True or false, with one line of reason each. (A) Squeezing the pipe raises the pressure inside the squeeze. (B) The fluid speeds up because something pushed it harder. (C) Energy is lost at the throat.
5. The three standpipes read the pressure beneath them. (A) Which column is the shortest, and why? (B) The third column stands as tall as the first. What does that tell you that the second one on its own does not?
6. Write "narrow → fast → low pressure" out as a chain, and name the conservation law behind each arrow.

i) (A) **False**, and this is the one worth arguing about: the throat is the fastest part of the flow and therefore the *lowest*-pressure part. (B) **True**, in the only sense available: the pressure *behind* it is higher than the pressure ahead of it, and that difference is the net force that accelerates it into the throat. (C) **False** — it is converted, not lost
ii) (A) the one over the **throat**, because that is where the flow is fastest and so the pressure is lowest. (B) that nothing was **lost**, only traded: the fluid gets its pressure back as soon as it slows down again a short second column alone would be consistent with energy disappearing; the tall third column rules that out
iii) narrow → fast is **mass** conservation (continuity); fast → low pressure is **energy** conservation (Bernoulli)

Bernoulli's equation — energy conservation for a fluid

Now make that bookkeeping exact. Take the work–energy theorem you used for blocks and carts in Module 4, apply it to a lump of fluid of volume V, and divide every term by V. Gravity does work on the lump, and so does the surrounding fluid, which pushes it along with a force PA over a distance — a work of $P \times$ volume. Nothing new goes in; what comes out is Bernoulli's equation.

7. (A) Write the work–energy theorem for the lump as $\tfrac{1}{2}mv_f^2 - \tfrac{1}{2}mv_i^2 = -[\,\Delta\text{PE} + \Delta P \cdot V\,]$, and say which work the last term is. (B) Divide every term by V and say what m/V becomes. (C) Collect the i's on one side and the f's on the other, and write out Bernoulli's equation.

8. Name each of the three terms, and say what quantity each one is per unit of what. Which of the three ran the whole of Note 1, and what has to be true of the flow for the other two to drop out?

9. A horizontal pipe narrows. The water moves at 2.0 m/s in the wide part, where the pressure is 150 kPa, and the narrow part has **half** the area. Take $\rho = 1000$ kg/m³ and find the pressure in the narrow part.

i) (A) $\Delta P \cdot V$ is the **work done by the surrounding fluid**, pressure times volume. (B) dividing by V turns m/V into **ρ**, and every term is now an energy *per unit volume*. (C) $\mathbf{\frac{1}{2}\rho v_i^2 + \rho g h_i + P_i = \frac{1}{2}\rho v_f^2 + \rho g h_f + P_f}$

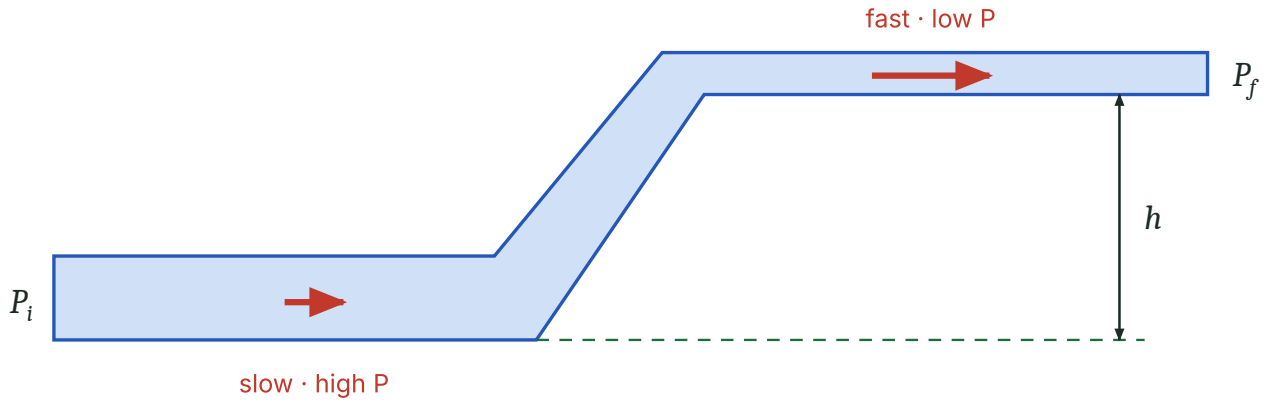
ii) $\frac{1}{2}\rho v^2$ is the **kinetic** term, ρgh the **potential** term, P the **pressure** term, each an energy per unit volume — which is why they can be added and why they all come out in pascals. **ρgh** ran Note 1: switch the flow off, $v_i = v_f = 0$, and Bernoulli collapses to the depth result. Fluid statics is this equation at rest

iii) continuity first, Bernoulli second. $A_n = \frac{1}{2}A_w$ gives $v_n = 2v_w = 4.0$ m/s; the pipe is horizontal so the ρgh terms cancel, leaving $P_n = P_w + \frac{1}{2}\rho(v_w^2 - v_n^2) = 150\,000 + \frac{1}{2}(1000)(4.0 - 16.0) =$ **144 kPa**

6 kPa out of 150 — the v^2 term only becomes dramatic when the speeds get large, which is why this effect is spectacular for aircraft and subtle for plumbing

When the pipe also rises

So far the pipe has been flat and the ρgh terms have cancelled. Let them not cancel. Here the pipe **narrows** and **climbs** at the same time, so the fluid gains kinetic energy *and* gains potential energy — and both bills land on the same account. Grouped the way it is easiest to read the signs off, Bernoulli says **ΔKE = −[ΔPE + ΔP]**, all three per unit volume.



both the speed and the height go up — so the pressure has to fall twice over

The narrow upper section is drawn at **half** the area of the lower one, so continuity puts the fluid there at **twice** the speed — and the upper arrow is drawn twice as long. The dashed line is the datum the height h is measured from; only the *difference* in height matters, so you may put it wherever is convenient.

10. For the pipe in the figure, which narrows and climbs: (A) what is the sign of ΔKE, and why? (B) what is the sign of ΔPE? (C) what do those two force the sign of ΔP to be, and roughly how big?

11. Compare it with the same pipe kept flat. Is the pressure drop larger or smaller here, and which term is responsible for the difference?

12. Name one everyday consequence of this, and say which of the three Bernoulli terms causes it.

    i) (A) **positive** — half the area, so continuity doubles the speed. (B) **positive** — it climbs. (C) for the bracket to come out negative overall, ΔP must be **negative, and big enough to outweigh the climb**: the pressure is paying for the speed *and* for the height

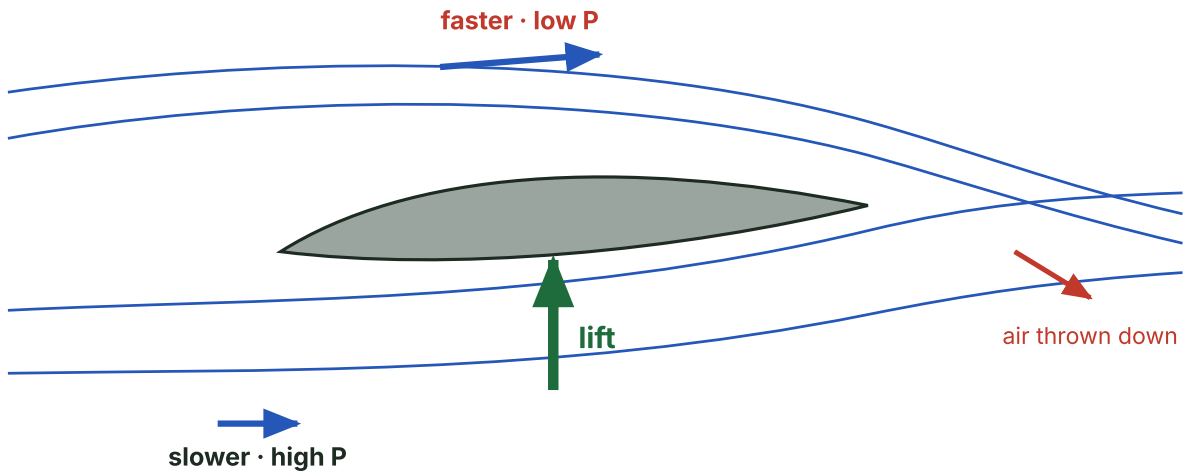
    ii) **larger** here. With $h = 0$ the pressure only had to cover the speed-up; here it covers the climb as well, so P_f at the top is much lower than it would be at the same speed on the flat. The **ρ*gh*** term is the difference

    iii) the top floor of a building has weaker water pressure than the ground floor, and a pump has to do work to fill a roof tank — ρ*gh* again same term, same note, now with the flow switched on

The wing

A wing is the famous case, and it is worth getting right rather than getting slick. The air passing over the upper surface moves **faster** than the air passing underneath; by Bernoulli, faster air is at **lower**

pressure; so the pressure underneath exceeds the pressure above, and that difference, acting over the whole area of the wing, is **lift**. That much is right. The usual explanation of *why* the upper air is faster is not.



The streamlines over the top are drawn **crowded** and the ones underneath **spread out** — crowded streamlines are fast-moving fluid, which is the same continuity idea as the narrow pipe. The red arrow at the trailing edge is the other half of the story.

13. State the Bernoulli account of lift in two sentences: what is faster, what is therefore lower, and what the difference acts over.

14. "The air over the top has further to go and has to meet the air from underneath at the trailing edge." Is that right? Give **two** separate pieces of evidence for your answer.

15. If that is not the reason, what *does* set up the speed difference? Say how the Newton account and the Bernoulli account are related.

i) the air over the upper surface moves faster than the air underneath, so by Bernoulli the pressure above is lower than the pressure below. That pressure difference, acting over the whole wing area, is the lift

ii) **No** — and this story is extremely common, so it is worth knowing why. First, nothing requires two parcels that separated at the leading edge to arrive at the trailing edge together, and measurements show they do not: the air over the top arrives *sooner*, not merely faster. Second, a wing that was symmetric top and bottom, or one flying upside down, would generate no lift on that account, and both fly perfectly well

iii) the wing's shape and its **angle to the oncoming air deflect the air downward**; by Newton's third law, pushing air down pushes the wing up — the red arrow in the figure. That same deflection

is what sets up the faster flow over the upper surface, and Bernoulli then says faster flow is at lower pressure two descriptions, one wing: *air pushed down* and *low pressure on top* are the same lift counted twice, not two competing causes

FOR PHY 151 — WITH CALCULUS

Bernoulli as an integral along a streamline. For steady flow of an inviscid, incompressible fluid, Newton's second law applied to a fluid element along its own streamline is Euler's equation, $\rho v\, \mathrm{d}v/\mathrm{d}s = -\mathrm{d}P/\mathrm{d}s - \rho g\, \mathrm{d}h/\mathrm{d}s$, with s measuring distance along the streamline. Multiply through by ds and integrate from one point to another: every term is an exact differential, so the integral collapses to $\tfrac{1}{2}\rho v^2 + P + \rho g h = \text{constant}$. Written this way the assumptions are visible rather than buried — **steady** (no $\partial v/\partial t$ term), **inviscid** (no friction term to integrate), **incompressible** (ρ comes out of the integral), and **along one streamline** (the constant may differ from streamline to streamline). Every failure of Bernoulli you will meet is one of those four being false.

16. A tank of water has a small hole a depth d below the surface. Apply Bernoulli from the surface to the hole, argue that both pressures are atmospheric and that the surface speed is negligible, and obtain **Torricelli's result**.

17. That is the same speed as an object dropped from height d. Say why, in terms of which Bernoulli terms survived.

18. The tank drains, so d falls. Write $\mathrm{d}d/\mathrm{d}t$ in terms of the hole area a, the tank area A and d, and say what shape of curve $d(t)$ is.

    i) both points are open to the air, so $P_\mathrm{i} = P_\mathrm{f}$ and the pressure terms cancel; the tank is wide compared with the hole, so continuity makes the surface speed negligible. What is left is $\rho g d = \tfrac{1}{2}\rho v^2$, so $\boldsymbol{v = \sqrt{(2gd)}}$

    ii) because the only two terms that survived are the **kinetic** and the **potential** ones — which is the energy statement for free fall, written per unit volume instead of per unit mass. Same equation, same answer

    iii) the tank level falls at the rate the hole empties it: $A\, \mathrm{d}d/\mathrm{d}t = -av$, so $\boldsymbol{\mathrm{d}d/\mathrm{d}t = -(a/A)\sqrt{(2gd)}}$. Separating gives $\sqrt{d}$ falling linearly in time, so $d(t)$ is a **parabola** — the level drops quickly at first and ever more slowly which is why a draining tank takes far longer to finish than to start

Where this sits. Continuity and Bernoulli close Module 6. The method of Note 1 — weighing a sample twice — is what you take into the **Buoyancy Lab**; the pressure-versus-speed result from this note is what makes the flow demonstrations in class work the way they do.

BACK TO YOUR PREDICTION

The pressure in the narrow section is **lower**. Continuity forces the water to speed up there, and the extra kinetic energy is paid for out of pressure. If you predicted "higher" you are in good company — the squeeze feels like it should pressurise the water — but the standpipe over the throat is the short one, and the pressure comes back up again once the pipe widens and the water slows.

Check yourself

1. Water flows at 3.0 m/s through a pipe of cross-sectional area 0.020 m². The pipe narrows to 0.0050 m². What is the speed in the narrow section, in m/s?

_____ m/s

2. In that narrow section, compared with the wide section, the pressure is

- ○ higher, because the same water is forced through a smaller opening
- ○ lower, because the water is moving faster there
- ○ the same, because the pipe is horizontal
- ○ impossible to say without the density

3. Which term of Bernoulli's equation is the one that carries over unchanged from the fluid statics of Note 1?

- ○ $\frac{1}{2}\rho v^2$
- ○ $\rho g h$
- ○ P
- ○ none of them — statics is a separate result

4. A pipe narrows *and* rises. Taking $\Delta KE = -[\Delta PE + \Delta P]$, the pressure change ΔP is

- ○ positive, and small
- ○ negative, and smaller in size than for the same pipe kept flat
- ○ negative, and larger in size than for the same pipe kept flat
- ○ zero, because the two gains cancel

Your notes

BEFORE THE VIDEO

Guided notes — blank (PDF)

print or open on a tablet; record your prediction, then draw the figures and work the problems as the video goes through them.

AFTER THE MODULE

Completed notes (PDF)

released when Module 6 closes.

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